ABSTRACT

Recent advancements in Multimodal Artificial Intelligence (AI) have significantly improved the ability of intelligent systems to understand, interpret, and reason using information from multiple modalities such as images and text. This seminar presents a comprehensive review of four recent research works that collectively enhance multimodal AI through efficient reasoning, adaptive computation, visual perception, and explainable artificial intelligence. The Mirage framework introduces the concept of Machine Mental Imagery, enabling Vision-Language Models (VLMs) to perform multimodal reasoning using latent visual tokens instead of explicit image generation, thereby improving reasoning efficiency while reducing computational overhead. ZipVL addresses the computational and memory challenges of large Vision-Language Models by dynamically selecting important visual tokens, accelerating inference through adaptive token sparsity without significantly affecting model accuracy. The AURORA framework enhances visual reasoning by introducing perception tokens that represent intermediate visual information such as depth maps and object locations, enabling models to perform more accurate spatial reasoning, object counting, and perception-based decision making. In addition, MaNtLE improves the transparency and interpretability of machine learning models by generating faithful natural language explanations that clearly describe the reasoning behind model predictions using a model-agnostic approach. Together, these research contributions demonstrate how latent reasoning, perception-aware learning, efficient computation, and explainable AI can be integrated to build more accurate, efficient, reliable, and trustworthy multimodal intelligent systems capable of addressing complex real-world applications across computer vision, robotics, healthcare, autonomous systems, and human-computer interaction.

Keywords— Multimodal AI, Vision-Language Models, Multimodal Reasoning, Latent Visual Tokens, Dynamic Token Sparsity, Perception Tokens, Explainable Artificial Intelligence (XAI), Computer Vision, Deep Learning, Artificial Intelligence.